

University of Washington

Department of Mathematics

Final Examination

MATH 394: Probability I

Summer 2026 — Instructor: Arman Jahangiri

Date: August 21, 2026

Answering time: 60 minutes

Total points: 60

Name: _____

Student ID: _____

Signature: _____

Instructions

- This examination contains **seven problems**. Verify that your copy is complete.
- Show all work and justify your answers. Unsupported answers may receive little or no credit.
- Clearly define any events or random variables that you introduce.
- The estimated times are provided only as a guide for pacing.
- Do not remove pages from the examination booklet.

Permitted materials

Examinations are closed-book and closed-notes. Students may use:

- one handwritten 8.5×11 inch reference sheet (front and back), and
- a non-programmable non-graphing calculator capable of standard exponential and logarithmic computations (e.g. TI-30X IIS).

No other electronic devices, calculators, communication devices, or external resources may be used during examinations.

Problem	Score
1	/16
2	/8
3	/7
4	/7
5	/8
6	/8
7	/6
Total	/60

By signing above, you affirm that the work is your own and that you followed the examination rules.

Problem 1.**Estimated time: 13 min****16 points**

Let the joint probability density function of random variables X and Y be given by

$$f(x, y) = \begin{cases} x^2 e^{-x(y+1)}, & x \geq 0, \quad y \geq 0, \\ 0, & \text{elsewhere.} \end{cases}$$

The marginal probability density functions are

$$f_X(x) = \begin{cases} x e^{-x}, & x \geq 0, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$f_Y(y) = \begin{cases} \frac{2}{(1+y)^3}, & y \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Calculate

$$\text{Cov}(X, Y).$$

(4 points)

(b) Calculate

$$\text{Cov}(3X + 1, 2X - 4).$$

(3 points)

(c) Are X and Y independent? Justify your answer.

(2 points)

(d) Calculate the moment-generating function of X , namely

$$M_X(t) = E(e^{tX}).$$

(5 points)

(e) Using $M_X(t)$, calculate $E(X)$.

(2 points)

Problem 2.**Estimated time: 9 min****8 points**

Let X and Y be two positive independent continuous random variables with probability density functions $f_1(x)$ and $f_2(y)$, respectively.

Find the probability density function of

$$U = \frac{X}{Y}.$$

Hint: Let $V = X$. Find the joint probability density function of U and V (Jacobian method), and then calculate the marginal probability density function of U .

Problem 3.**Estimated time: 5 min****7 points**

Let X and Y be i.i.d. positive random variables. Assume that all expectations appearing below exist and are finite.

For each part below, fill in the appropriate equality or inequality symbol.

Write $=$ if the two sides are always equal, $\leq$ if the left-hand side is less than or equal to the right-hand side but not necessarily equal, and similarly for $\geq$. If no relation holds in general, write $?$.

Justify each answer.

(a)

$$E(\log X) \quad ______ \quad \log(E(X)).$$

(3 points)

(b)

$$P(X \leq Y) \quad ______ \quad P(X \geq Y).$$

(2 points)

(c)

$$E(XY) \quad ______ \quad \sqrt{E(X^2)E(Y^2)}.$$

(2 points)

Problem 4.**Estimated time: 8 min****7 points**

Suppose

$$X_1, \dots, X_n$$

are independent Bernoulli random variables with unknown success probability p , and define

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Let

$$\varepsilon > 0 \quad \text{and} \quad 0 < \alpha < 1.$$

Using Chebyshev's inequality, derive a condition on the sample size n that guarantees

$$P(|\hat{p} - p| < \varepsilon) \geq 1 - \alpha$$

for every

$$0 < p < 1.$$

Problem 5.**Estimated time: 7 min****8 points**

Let X and Y be independent exponential random variables with common parameter $\lambda > 0$:

$$X, Y \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(\lambda),$$

with probability density function

$$f(w) = \begin{cases} \lambda e^{-\lambda w}, & w > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Define

$$T = X + Y.$$

Using the convolution formula

$$f_T(t) = \int_{-\infty}^{\infty} f_X(x) f_Y(t - x) dx,$$

find the probability density function of T , including its support.

Problem 6.**Estimated time: 5 min****8 points**

Let X, Y, Z be jointly continuous with joint probability density function

$$f_{X,Y,Z}(x, y, z) = \begin{cases} x^2 e^{-x(1+y+z)}, & x, y, z > 0, \\ 0, & \text{otherwise.} \end{cases}$$

The marginal probability density functions are

$$f_X(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$f_Y(y) = \begin{cases} \frac{1}{(1+y)^2}, & y > 0, \\ 0, & \text{otherwise,} \end{cases} \quad f_Z(z) = \begin{cases} \frac{1}{(1+z)^2}, & z > 0, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Are X, Y, Z pairwise independent? Justify your answer. **(5 points)**
- (b) Are X, Y, Z mutually independent? Justify your answer. **(3 points)**
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Problem 7.**Estimated time: 4 min****6 points**

For each part below, select the **correct statement** of the indicated theorem. Only one choice is correct in each part.

(a) Weak Law of Large Numbers (WLLN).

Let $X_1, X_2, \dots$ be i.i.d. random variables with finite mean

$$E(X_i) = \mu$$

and finite variance

$$\text{Var}(X_i) = \sigma^2 > 0.$$

Let

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Which of the following is the conclusion of the Weak Law of Large Numbers?

(2 points)

(A) For every $\varepsilon > 0$,

$$P(|\bar{X}_n - \mu| > \varepsilon) = 0 \quad \text{for every } n.$$

(B) For every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| > \varepsilon) = 0.$$

(C)

$$P\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1.$$

(D) For every $z \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} P\left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq z\right) = \Phi(z).$$

(b) Strong Law of Large Numbers (SLLN).

Let $X_1, X_2, \dots$ be i.i.d. random variables satisfying the same assumptions as in the Weak Law of Large Numbers, with

$$E(X_i) = \mu$$

and finite variance

$$\text{Var}(X_i) = \sigma^2 > 0.$$

Which of the following is the conclusion of the Strong Law of Large Numbers?

(2 points)

(A)

$$P\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1.$$

(B) For every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| > \varepsilon) = 0.$$

(C)

$$\lim_{n \rightarrow \infty} P(\bar{X}_n = \mu) = 1.$$

(D)

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

(c) **Central Limit Theorem (CLT).**

Let $X_1, X_2, \dots$ be i.i.d. random variables with

$$E(X_i) = \mu, \quad \text{Var}(X_i) = 0 < \sigma^2 < \infty.$$

Which of the following is the conclusion of the Central Limit Theorem?

(2 points)

(A)

$$\bar{X}_n \xrightarrow{P} N(\mu, \sigma^2).$$

(B)

$$P\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1.$$

(C) For every $z \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} P\left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq z\right) = \Phi(z).$$

(D) For every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| > \varepsilon) = 0.$$
